Rishil Patel

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Education

Northeastern University, Boston, MA May 2027

Bachelor of Science in Computer Engineering and Computer Science, GPA 3.98

- **Coursework:** Fundamentals of Software Engineering, Algorithms and Data, Intro to Artificial Intelligence, Capstone, Object-Oriented Design, Computer Systems, Linear Systems, Circuits & Signals, Embedded Design, Probability and Statistics, Differential Equations
- **Activities:** Capture the Flag NU, Trivia Club, Service Learning Program

Experience

United Electronic Industries, an AMETEK Company, Norwood, MA January – June 2026

Software Engineering Co-op

- Designed a Python PyTest regression framework for UEI data acquisition and I/O boards. Wrote regression tests for 20 different boards.
  - Wrote tests for analog and digital I/O, and the CAN and MIL-STD-1553 communication protocols.
  - Replaced static device configuration with network-based hardware discovery, reducing setup time to under a minute and eliminating manual configuration errors.
- Fixed 30+ customer and internally-reported issues across UEI's C, C++, and Python APIs.
- Refactored the SWIG rules responsible for wrapping UEI's C++ API to generate its Python API.

E Ink Corporation, Billerica, MA January – August 2025

Hardware Interface Development Co-op

- Designed and implemented an automated quality control system to enable high-volume inspection of E Ink panels
  - Developed Python-based software to capture and analyze optical data
  - Incorporated calibration procedures (flat-field correction, color reference adjustment) to ensure measurement accuracy and consistency
  - Enabled 100% panel inspection and scalability to over 4 million units annually
- Developed Python interfaces to integrate spectrometers and other devices into an E Ink panel measurement framework

Engineering Projects

Computer Vision and Multimodal Lift Analysis for Injury Monitoring and Prevention (CLAMP) July 2026 – Present

- Collaborating in a 5-person capstone team advised by Northeastern's Augmented Cognition Lab to build a computer vision system detecting fatigue-driven form breakdown in the squat, bench press, and deadlift exercises
- Leading integration of a vision-language model with real-time egocentric video and pose estimation from dual Meta Project Aria glasses, worn by the lifter and a spotter, to detect improper form and flag compensatory movement
- Contributing to a multimodal sensing pipeline (dual egocentric video, barbell-mounted IMU, MudraLink wrist sensor) feeding a web app that delivers real-time voice warnings during a lift and post-set analytical feedback

Adaptive Home Safety Monitoring System September 2025 – December 2025

- Collaborated in a 4-person team to develop an assistive door monitoring system for children with special needs, guided by family input
- Designed a multi-sensor approach combining reed switches, accelerometers, and ultrasonic sensors to enhance reliability
- Developed an ESP32-based system to provide real-time door status updates through SMS and email notifications
- Implemented fail-safe features including tamper-resistant mounting and a dual-power system with primary and redundant backup batteries

Technical Skills

Languages: Python (PyTest, NumPy, pandas, matplotlib), C, C++, Java, MATLAB, TypeScript

Developer Tools: Git/GitHub, SVN, Linux, LTspice, AI Tools (Claude Code, GitHub Copilot, Cursor)

Hardware & Lab: Digital multimeter, oscilloscope, waveform generator, logic analyzer, soldering, Raspberry Pi